

Dynamic Programming (Knapsack & LCS)

DOMAIN: DATA STRUCTURES & ALGORITHMS TIMELINE: WEEK 6

RELEASE: BATCH 2026

1. Project Scope & Description

Build memoized and tabulated solutions for 0/1 Knapsack and Longest Common Subsequence.

2. Key Technical Requirements

- 1. Write top-down memoization arrays controllers.
- 2. Declare bottom-up tabulation matrix indices loops.
- 3. Prove time complexity optimizations from recursions.

3. Assignment Submission Guidelines

%^a GitHub Repository: Create a public repository on GitHub. Push your source code files regularly and write a professional README.md with screenshots and setup instructions.

%^a LinkedIn Sharing: Write a brief post sharing what you built, what technologies you used, and what you learned. Tag @Jobify and attach a screenshot/video.

%^a Portal Submission: Paste both your public GitHub repository URL and your LinkedIn post URL inside the student assignment submission panel.

& p CRITICAL SUBMISSION POLICIES:

Verify that your GitHub repository visibility is set to public. Private links cannot be analyzed by our automated evaluation engines and may lead to credit deductions.